Preview: y2007_q03 (v2)

Comparing sets of letters: each different letter counts once; check them all

Part A: Each question is worth 3 points

1. How many letters do the words ELEPHANT and NUMBER have in common?

A. 4 ***B.*** 3 ***C.*** 2 ***D.*** 1 ***E.*** 0

2. How many letters do the words ELEPHANT and SPIDER have in common?

A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

3. How many different letters are used to write the word CHOCOLATE?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

4. How many different letters are used to write the word WINDOW?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

Part B: Each question is worth 4 points

5. Mia writes the word MATHEMATICS. Then Mia crosses out every letter that also appears in the word TRIANGLE.
How many different letters are not crossed out?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

6. Leo writes the word SCHOOL. Then Leo crosses out every letter that also appears in the word KANGAROO.
How many different letters are not crossed out?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

7. Which word has exactly 3 different letters in common with PENCIL?

A. DRAGON ***B.*** MONKEY ***C.*** SPIDER ***D.*** TEACHER ***E.*** KANGAROO

8. Which word has exactly 2 different letters in common with CHOCOLATE?

A. PROBLEM ***B.*** PLANET ***C.*** CASTLE ***D.*** SQUARE ***E.*** SPIDER

Part C: Each question is worth 5 points

9. Mia has letter cards and wants to show which letters are used in the words MATHEMATICS and PLANET. Mia puts down one card for each different letter used in either word.
How many cards does Mia put down?

A. 17 ***B.*** 14 ***C.*** 12 ***D.*** 11 ***E.*** 10

10. Mia has letter cards and wants to show which letters are used in the words CHOCOLATE and RAINBOW. Mia puts down one card for each different letter used in either word.
How many cards does Mia put down?

A. 12 ***B.*** 11 ***C.*** 10 ***D.*** 9 ***E.*** 8

Answer Key

1.	**C**	**2.**	**E**	**3.**	**B**	**4.**	**A**	**5.**	**E**	**6.**	**B**
7.	**C**	**8.**	**D**	**9.**	**D**	**10.**	**A**				

Solutions

1. C (2) — Check each different letter of ELEPHANT: only E and N are also in NUMBER, so 2. A letter that appears twice still counts once. [2007 Q3 · common_letters]

2. E (2) — Check each different letter of ELEPHANT: only E and P are also in SPIDER, so 2. A letter that appears twice still counts once. [2007 Q3 · common_letters]

3. B (7) — The letters are A, C, E, H, L, O and T: 7 different letters (CHOCOLATE has 9 letters in all). [2007 Q3 · different_letters]

4. A (5) — The letters are D, I, N, O and W: 5 different letters (WINDOW has 6 letters in all). [2007 Q3 · different_letters]

5. E (4) — Crossed out: A, E, I and T (they are in TRIANGLE). Left: C, H, M and S, so 4 different letters (a letter that is left twice counts once). [2007 Q3 · only_in_first]

6. B (4) — Crossed out: O (they are in KANGAROO). Left: C, H, L and S, so 4 different letters. [2007 Q3 · only_in_first]

7. C (SPIDER) — SPIDER shares E, I and P with PENCIL. KANGAROO shares 1; DRAGON shares 1; MONKEY shares 2; TEACHER shares 2. [2007 Q3 · which_word]

8. D (SQUARE) — SQUARE shares A and E with CHOCOLATE. PROBLEM shares 3; PLANET shares 4; SPIDER shares 1; CASTLE shares 5. [2007 Q3 · which_word]

9. D (11) — MATHEMATICS uses 8 different letters and PLANET uses 6. A, E and T are in both, so count them once: 8 + 6 – 3 = 11. [2007 Q3 · letter_cards]

10. A (12) — CHOCOLATE uses 7 different letters and RAINBOW uses 7. A and O are in both, so count them once: 7 + 7 – 2 = 12. [2007 Q3 · letter_cards]